Comparative Study and Simulative Analysis of High Voltage and Low Voltage Distribution Systems for Rural Agricultural Loads

Simulation Using Etap Software

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***Abstract*—In India, the total transmission and distribution losses are as high as 22.77 percentage and the theft of the electricity is also a huge issue. This scenario is not only in India but it is also there in many developing countries. To solve the issues related to high losses in the distribution stage of power system for agricultural loads of rural areas, High Voltage Distribution System (HVDS) has been proposed. There are basically four objectives of this paper. First is to present the proper literature review of the IEEE-published papers which are listed under HVDS in the chronological order of their publication year which would be helpful to new researchers for understanding the step-by-step development in the field of HVDS. Second is to present the comparative study of existing Low Voltage Distribution System (LVDS) and the proposed HVDS on the basis of the analysis which was performed. Third is to present the simulative analysis of both the systems on the basis of the simulation which was made using etap software. Using etap for this analysis makes this paper distinctive than all the other research papers which are reviewed in this paper. Fourth is to present a cost-analysis of the proposed HVDS system. Overall, this paper is also a small and humble approach in the direction of realizing the schemes like DDUGJY (Deendayal Upadhyaya Gram Jyoti Yojana) under rural electrification plan of the Ministry of power of the republic of India.**

Keywords-HVDS; LVDS; distribution systems; etap; cost analysis; simulation; literature review; comparitive study

I. INTRODUCTION

Electrical distribution system basically comprises the parts of electrical power system between the sub transmission system and the consumers service. This includes distribution substations, distribution transformers, primary distribution feeders, secondary circuits, services to consumer, and appropriate protective and control devices. The Indian power distribution for agricultural loads consists of largely three phase 11 kV main distribution feeders, with three phase spur lines and 11/0.44 KV three phase distribution transformers. This is basically a low voltage distribution system (LVDS). India's total installed capacity is 315.42 GW as of February 2017. Among which 18.17% is agriculture consumption [1]. Agricultural consumption is mainly due to induction motor pumps which extract ground water for giving it to crops. Total losses in transmission and distribution are 22.77% which is very high compare to developed countries. The losses that occur during the generation of power can be technically defined whereas the transmission and distribution losses cannot be precisely evaluated from the sending end data. This suggests that there is significant involvement of non-technical parameters in transmission and distribution of electricity [2]. The losses include both technical and non-technical losses. Technical losses include generation losses and the losses due to the current flowing in the electrical network such as line losses, copper and iron losses of the transformer. Technical losses are normally 22.5%, and directly depend on the network characteristics and the mode of operation. Fixed Technical losses consists of corona losses, leakage current losses, dielectric losses etc. and variable technical losses consists of Joule losses in lines in each voltage level, impedance losses, losses caused by contact resistance etc. The non-technical losses, also known as commercial losses, are those related to unmetered supplies, incorrect billing, untimely billing, wrong tariff, defective meters and energy thefts. The energy theft may occur when consumers tamper with the metering, or connive with the utility personnel to make illegal connections. The prevention of the theft which is committed by the people who are having agricultural connections in rural areas is one of the biggest challenges to power utilities of India. There are many steps which utility can take to reduce losses in distribution lines like, the use of proper jointing techniques, and keeping the number of the joints to a minimum, feeding heavy consumers directly from the feeders, proper load management and load balancing, use of electronic meters which are accurate and tamper-proof, improving power factor by adding shunt capacitors etc. But these solutions are to be performed on LVDS system. There is another solution which is to convert the distribution system into HVDS. This can elude the need of long 440 V lines which are necessary in case of existing LVDS systems. Conversion of LVDS into HVDS has been done by some private companies like SEALWEL in India [3]. It has been undertaken by some state electricity boards like Gujarat, Andhra Pradesh, Karnataka etc. But that is still not on a very large scale. One of the many reasons behind this can be that it needs proper cost analysis to know whether the money

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saved by reducing the losses over the years after installing, is higher than the cost of replacing existing LVDS with HVDS or not and if it is, then what is the payback period. After all this analysis, if it is found economical then and only then it is performed. After this introduction, the paper is distributed in 4 sections, first is literature review which is followed by comparative study, simulative analysis and cost analysis.

II. Literature Review

Proper literature review of the IEEE-published papers which are listed under HVDS was performed and it is presented in the chronological order of their publication year which would be helpful to researchers for understanding the step-by-step development in the field of HVDS.

Paper [4] gives basics of HVDS for the loads which are having low power factor and are scattered in large area. It gives equations for load flow analysis of HVDS as well as it also gives the algorithm for the same. An illustrative example of a practical radial distribution system is provided to illustrate the effectiveness of HVDS.

Paper [5] gives information about the concepts of HVDS. It discuss about the types of HVDS, advantages of HVDS, comparison between HVDS and LVDS. It gives analysis of 33/11 KV Kovour substation Mancherry feeder using simulation. Various plots of the analysis are also presented in the paper.

Paper [6] discuss about Voltage Profile Improvement through HVDS. Authors have first given information about the losses in distribution system, their reasons and remedies. Then HVDS system as well as existing system is simulated using PSCAD and the results of both the simulations are compared. Cost analysis is also done.

Paper [7] gives idea about the method of implementing HVDS replacing the existing LVDS. The analysis is carried out using CYMDIST. The flowchart of the steps for the loss reduction is also given.

Papers [8], [9] propose a different kind of HVDS where distributor phase voltage is increased to 350V and at the consumer premises, it is stepped down to 230V using special voltage regulator device. The working of this methodology is illustrated with the setup of lamps. The analysis of the same is also done in MATLAB.

Partially insulated Covered conductor and fully insulated ABC (Aerial Bunched Cable) systems are proposed in paper [10]. which can permit HVDS lines to pass through narrow alleyways and slums without allowing illegal tapping.

Paper [11] tries to determine voltage profile of HVDS systems using forward sweep method and it also includes feedback of consumers who have moved to HVDS from LVDS.

In the HVDS which are having booster transformers, to overcome the effects of transformer, a step-up power converter with fuzzy control has been proposed in paper [12].

III. Comparative Study of Proposed HVDS and Existing LVDS

The existing LVDS systems for rural agricultural loads are such that they are having 11/0.44 KV three phase distribution transformers and on the Low Tension (LT) side of the transformer, long 440V distributors are needed because the load density of such loads is generally low due to the reason that the farms are spread across huge areas. So the problem with LVDS for this type of loads is the losses due to long LT distributors. Another major problem is the theft of electricity. Because the LT distributors are very long, it increases the probability of theft. As a remedy to these issues with LVDS, this paper proposes HVDS for rural agricultural loads. This paper proposes that instead of having a DTR (distribution transformer) of high rating like 50-100 kVA to serve the load of 20-30 pump sets, 11 KV line can be brought to consumer's premises and then a small single phase, single pole mounted transformer of small rating like 5-15 kVA can be used to serve an agricultural load of induction motor pump set. In place of a single phase transformer, using a three phase transformer is also an option. If that is chosen, then 2-3 such loads can be served from one DTR. The system's aim is to achieve low LT lines or almost no LT lines. This system will basically serve 2 purposes, because the distribution is on higher voltage levels, the losses will reduce and in this system, LT line is extremely small so it is economical to get it insulated so that it reduces (almost removes) the probability of theft compare to the existing systems. Some disadvantages of LVDS, advantages of HVDS as well as comparison of both of the systems is presented below.

A. Disadvantages of LVDS

- Tail end voltages are not good.
- High probability of losses.
- Fuse blow outs and jumper cuts can happen frequently.
- Burn outs of motor are nearly twice in each cropping period of around 100 days.
- Frequent faults can cause DTR (distribution transformer) failure.
- In rural areas, replacement of failed DTRs can take time and due to that, the loss of standing crops will happen.
- LV line losses are 6 times beyond the targeted limit and 3 times beyond the maximum tolerable limit which is very far from the international norms [13].

B. Advantages of the Proposed HVDS

- Smaller size conductors can be used.
- Good voltage profile can be achieved.
- Losses can be reduced.
- Frequent fuse blowouts can be prevented.
- Transformer failures are almost nil.
- Two or three consumers get affected against 30 if a single DTR fails.
- Easy and quick replacement can be done.
- Unauthorized hooking of loads is impossible as LT lines are insulated and very short.
- System power factor can be improved.
- As only 2-3 pump sets are connected with each DTR, the consumers feel the responsibility and ownership.

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- High quality of power supply will yield customer satisfaction.
- Accidents because of touching of conductors is reduced due to the fact that the breaker trips at the substation since the line is at 11 KV potential.
- Additional generation capacity is not needed for giving new loads due to reduction in power drawals.
- High quality of supply since there is ideally no voltage drop.
- Failure will be minimal because of over loading as well as meddling of L.T lines is prevented.
- Single phase motors of good quality are available for the loads which are there for irrigation purpose.
- High yield of water because the pump set efficiency is on the higher side.
- Single phase motors are essentially 'capacitor start and capacitor run' so costly starters are not required.

C. *Comarison of LVDS and the Proposed HVDS*

- Unauthorized tapping of LV lines is simple and very much probable in LVDS whereas it is extremely difficult in HVDS.
- Power loss for transmission of equal load in LVDS and HVDS are of ratio 13:1 [13].
- Maximum permissible drop between distribution substation and customer premises is 10%. Hence compliance with Indian Electricity rule, 1956 is there in HVDS whereas in LVDS, that is not the case [14].
- About 75% of LV feeders have drops above 5% and that is the cause of high losses whereas in HVDS losses on LV line are negligible.
- The current for distribution of same power in LVDS is high and conductors can get loaded beyond loading limits. This can be prevented by HVDS.
- The monitoring of feeders in LVDS is difficult compared to HVDS.

IV. Simulative Analysis of Proposed HVDS and Existing LVDS Using Etap Software

A simulation of the existing LVDS system as well as of the proposed HVDS system was made using etap software. Using etap for this analysis makes this paper distinctive than all the papers which are reviewed. etap is used because it is a modern and very powerful tool for this kind of simulation. In the simulation, standard values were taken into consideration like 11 kV conductor was taken ACSR, 50 mm², 6 strands, 10 km long and 440V cable was taken PVC insulated, copper, 6 mm², 50 m long in case of LVDS and 5 m long in case of HVDS. Induction motor rating was taken as 4 kW (Although, it can vary in the different parts of the country according to the ground water level at local region). Induction motor was taken as an intermittent load with the average running period of 10 hours a day. DTR in the simulation of LVDS is of 100 kVA and in HVDS, each DTR is of small rating of 10 kVA. In LVDS, the loadings due to theft are taken into account that is why more loads are connected in LVDS than HVDS. Static load of 1 kW is also considered which represents the continuous lighting load of the consumer. Although it is to be noted that, in some states of the country, the lighting load is given a different connection than the induction motor pump connection. Power grid supply is connected to supply these loads. After making both the simulations, several tests were performed on both of them with the purpose of comparing their performance,

A. *Different Types of Load Flow Analysis*

Total 4 types of load flow analysis were performed on both the simulations which are namely, load flow analysis, optimal load flow analysis, unbalanced load flow analysis and time domain load flow analysis. The results of the load flow analysis on both the simulations is as shown in Fig. 1, and Fig. 2. It was observed that power grid block in the simulation has to supply 35.6 kW in case of LVDS whereas in the case of HVDS, it is reduced to 27.2 kW which clearly indicates that the power needed to supply the same legitimate loads is reduced by 23.59% in HVDS compare to LVDS. This decrease is due to 2 main reasons which are, the loads which were connected as the theft loads in LVDS are not there in HVDS and second reason is, due to very less length of LT, the technical losses like copper losses etc. are reduced. Similarly, the results of other types of the load flow analysis also showed the decrease in the losses.

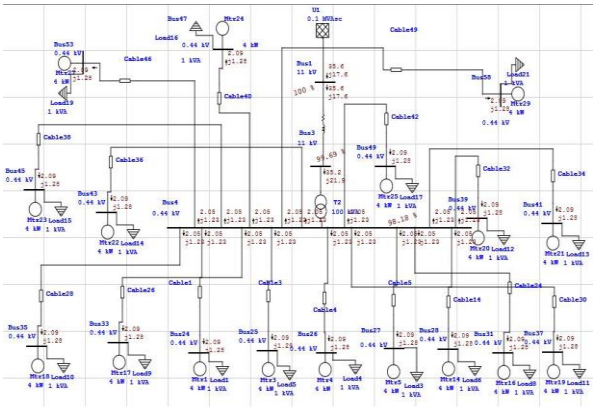


Figure 1. Load flow analysis of LVDS.

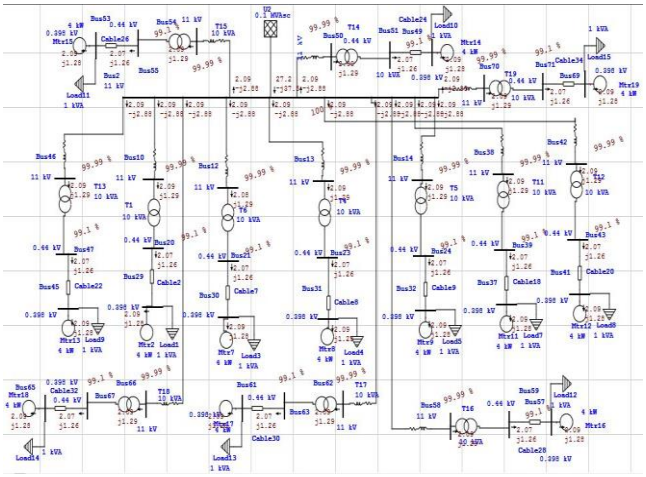


Figure 2. Load flow analysis of the proposed HVDS.

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B. *Short Circuit Analysis*

Although short circuit fault can be simulated at different buses or at the combination of more than one buses, the fault was simulated on main bus bar which is located after the DTR in case of LVDS and before the DTRs in case of HVDS looking from the supply side. L-G, L-L, L-L-G faults were simulated. The fault currents of LVDS in case of L-G and L-L-G were 0.232 kA and 0.465 kA respectively. In case of HVDS they were 0.005 kA and 0.004 kA. So there is significant decrease in fault currents in HVDS compare to LVDS which is advantageous for the system.

C. *Voltage Stability Analysis*

This test gives idea about the stability of voltage levels in the system. Stability can be disturbed by reactive power deficiency, unmitigated rotor angle instability etc. factors. From this test, stability at consumer bus bar was found 99.85% in case of HVDS and in case of LVDS it was 99.3%. So there is no huge change here due to HVDS but still the value is better than LVDS.

D. *Reliability Assessment*

This test basically gives the idea about the reliability of the system. Upon performing this assessment on both of them, average outage duration of pump set in case of LVDS is found 77.2 hours whenever failure happens. It is 29.1 hours in case of HVDS. So reliability of HVDS is better compare to LVDS

E. *Star Protection and Coordination*

This test is consisting of tools for performing equipment protection and protective device coordination. It provides informed and reliable recommendations regarding the feasibility of the devices under consideration. The results of this test on HVDS and LVDS are as shown in Fig. 3 and Fig. 4 respectively. The tree-like structure is showing the protection zones.

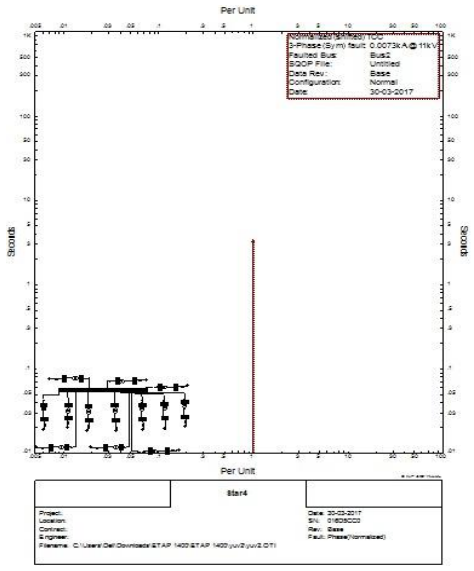


Figure 3. Star protection and coordination analysis Results for HVDS.

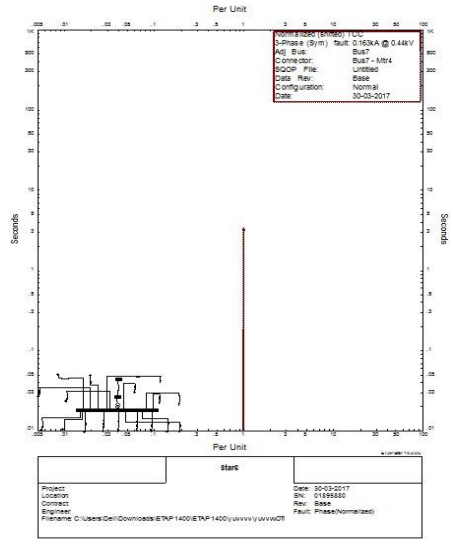


Figure 4. Star protection and coordination analysis Results for LVDS.

V. COST ANALYSIS

To check the feasibility of the proposed HVDS system considering the economic point of view, cost analysis was performed. In the analysis, existing LVDS systems are considered to be having 100 kVA DTRs which are getting replaced by 10 kVA transformers in the proposed system. The analysis was done on the scale of a single feeder coming out from the distribution substation. Considering the standard prices and quantities of transformers needed to supply the loads on a single feeder, the Table I is made. Another Table II is made considering the lengths and prices of HT as well as LT lines needed in both LVDS and the proposed HVDS. Here also, the base prices and base lengths are taken into account and the total prices are calculated in round figures in both the tables.

According to these results,

- Total Price of HVDS for 1 feeder = ₹500000 + ₹800000 = ₹1300000
- Total Price of LVDS for 1 feeder = ₹500000 + ₹575000 = ₹1075000

As seen clearly, the cost of the proposed HVDS is ₹225000 higher than existing LVDS. Which is around 120.93 % higher. This price is to be multiplied with the number of feeders in the system because this analysis is on a single feeder. So it is clear that higher capital investment would be required for HVDS. Now this proposed system will be feasible only if the cost invested initially is rewarded back by the savings in losses. To know this payback period, break-even analysis was performed. The kW values of savings in losses were taken from the etap simulation which was made. For converting this kW loss into money, in the calculation, load factor was taken 0.54 and loss factor was taken 0.33 [7]. And the kWh value is calculated considering the same thing as the etap simulation that the induction motor pump runs for 10 hours a day. Per unit price of supply is considered ₹3. After doing the calculation, per year cost

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saving due to less losses in HVDS was found ₹40178.57. The results of the break-even analysis which were obtained are as shown in Fig. 5. They show that the payback period is around 5.6 years.

TABLE I. TRANSFORMER PRICING

Types of Distribution systems	Transformer kVA rating	Quantity of transformers needed	Price of each unit of transformer (₹)	Total price of transformers (₹)
HVDS	10	50	10000	500000
LVDS	100	5	100000	500000

TABLE II. CABLE/CONDUCTOR PRICING

Types of Distribution systems	Length of HT (Price : ₹500/m) needed (m)	Length of LT (Price : ₹50/m) needed (m)	Price of whole length of HT (₹)	Price of whole length of LT (₹)	Total price (HT + LT price in ₹)
HVDS	1500	1000	750000	50000	800000
LVDS	1000	1500	500000	75000	575000

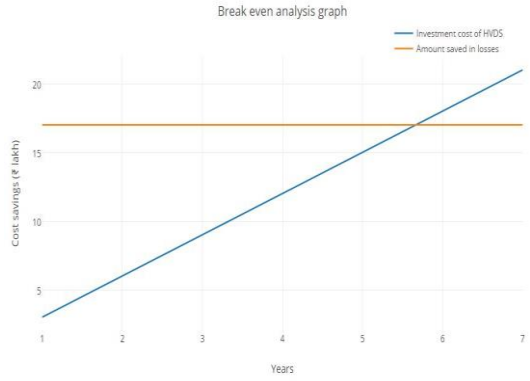


Figure 5. Results of break-even analysis.

VI. CONCLUSION

The literature review presented in this paper will be useful to new researchers for understanding the progress in the field of HVDS properly. The comparative analysis presented in this paper listed out the advantages of HVDS, disadvantages of LVDS and direct comparison between both the systems. Various tests and analysis performed on the etap simulation showed that the proposed HVDS is better than the current LVDS in various aspects like load flow analysis showed that power needed to supply the same legitimate loads is reduced by 23.59% in HVDS compare to LVDS, short circuit analysis showed that L-G and L-L-G fault currents are significantly reduced in HVDS compare to LVDS, Voltage stability test showed that the stability has been increased to 99.85% in HVDS from 99.3% of LVDS, reliability assessment test showed that the outage duration at the time of fault has been reduced to 29.1 in HVDS from 77.2 hours of LVDS. At the end, the cost analysis showed that that even though the capital investment is high in HVDS, payback period of that extra investment is not too long. It is around 5.6 years.

REFERENCES

[1] http://www.indiaenvironmentportal.org.in/files/file/growth_2015.pdf

[2] Soma Shekara Sreenadh Reddy Depuru, Lingfeng Wang, and Vijay Devabhaktuni, "Support Vector Machine Based Data Classification for Detection of Electricity Theft", IEEE Power Systems Conference and Exposition (PSCE), pp. 01-08, March 20th – 23rd, 2011.

[3] http://www.sealwel.com/hvds.html

[4] K. Amaresh, S. Sivanagaraju, and V. Sankar, "Minimization of Losses in Radial Distribution System by using HVDS", International Conference on Power Electronic, Drives and Energy Systems, 2006

[5] S.A. Sampath Kumar, V. Vasudaven, J. Antony, Madhu Sudhana Raju, L. Ramesh, "Minimization of Power Losses in Distribution System Through HVDS Concepts", Sustainable Energy and Intelligent Systems (SEISCON), 2011

[6] L. Ramesh, M. Madhusudhanaraju, S.P. Chowdhury, S. Chowdhury, "Voltage Profile Improvement through High Voltage Distribution System", Sustainable Energy and Intelligent Systems (SEISCON), 2011

[7] Md Sarwar, Anwar Shahzad Siddiqui, Zainul A. Jaffery, Imran Ahmad Quadri, "Techno-Economic Feasibility of HVDS Concept for Distribution Feeder Power Loss Minimisation", IEEE 5th India International Conference on Power Electronics (IICPE), 2012

[8] P Ravi Babu, Sushma B, Ashwin Kumar B, "HVDS approach for reducing the Technical and Non-technical losses to enhance the Electrical Distribution System performance", IEEE 5th India International Conference on Power Electronics (IICPE), 2012

[9] Dr P Ravi Babu, Sushma.B, "Operation and Control of Electrical Distribution System with Extra Voltage to minimize the Losses", International Conference on Power, Energy and Control (ICPEC), 2013

[10] H.K. Agarwal, K. Mukherjee, P. Barua, "Partially and Fully Insulated conductor systems for low and medium voltage over head distribution lines", IEEE 1st International Conference on Condition Assessment Techniques in Electrical Systems (CATCON), 2013

[11] K. Spandana, Varsha Reddy. A, "Restructuring of a Low Voltage Distribution System into a High Voltage Distribution System for an Improved Voltage and Power Loss Profile", International Conference and Utility Exhibition on Green Energy for Sustainable Development (ICUE), 2014

[12] Anusha Vadde, V S N Sitaramgupta V, "Enhancement of Distribution System Performance using HVAC Boost Converter and Fuzzy Controller", Annual IEEE India Conference (INDICON), 2015

[13] http://www.vijaielectricals.com/kb_hvds.html

[14] www.dgms.net/IErules1956.pdf

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